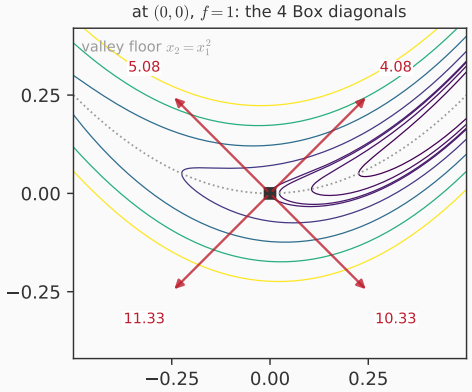
Optimization

3.2.5 Hooke–Jeeves method (pattern search)

José Firmino Aguilar Madeira

What if, instead of the diagonals, we try the axes?

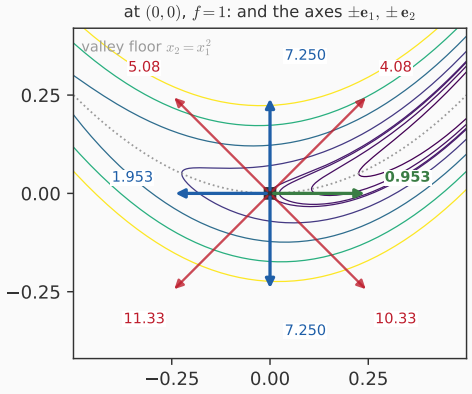
3.2.4 ended on Rosenbrock, at $(0, 0)$, $f = 1$: Box tests the 4 diagonals and none improves.



What if, instead of the diagonals, we try the axes?

3.2.4 ended on Rosenbrock, at $(0, 0)$, $f = 1$: Box tests the 4 diagonals and none improves.

But along the x_1 axis: $f(0.25,\ 0) = 0.953 < 1$.



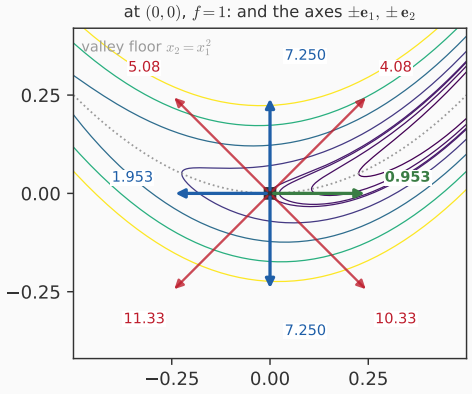
What if, instead of the diagonals, we try the axes?

3.2.4 ended on Rosenbrock, at $(0, 0)$, $f = 1$: Box tests the 4 diagonals and none improves.

But along the x_1 axis: $f(0.25,\ 0) = 0.953 < 1$.

Why not test $\pm\mathbf{e}_1, \ldots, \pm\mathbf{e}_n$?

n	Box: 2^n	axes: $2n$
2	4	4
10	1 024	20
20	$\approx 10^6$	40

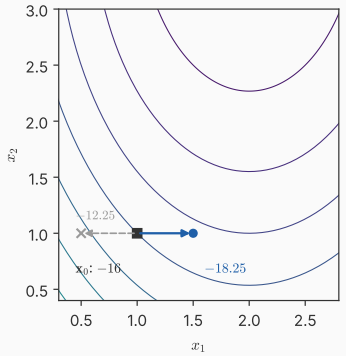


One exploration by hand

$f(x_1, x_2) = 3x_1^2 + x_2^2 - 12x_1 - 8x_2$ (minimum at $(2, 4)$, $f = -28$).
Base point $\mathbf{x}_0 = (1, 1)$, $f = -16$; steps
$\mathbf{\Delta} = (\Delta_1, \Delta_2) = (0.5,\ 0.5)$.

coordinate x_1, from $(1, 1)$

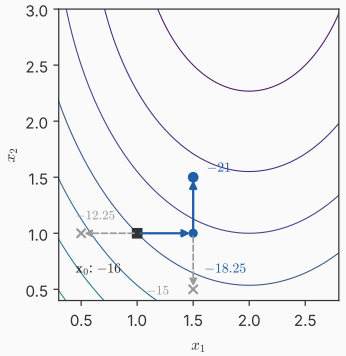
$(x_1 + \Delta_1,\ x_2)$	$= (1.5,\ 1)$	$f^+ = -18.25$
$(x_1,\ x_2)$	$= (1,\ 1)$	$f = -16$
$(x_1 - \Delta_1,\ x_2)$	$= (0.5,\ 1)$	$f^- = -12.25$



One exploration by hand

$f(x_1, x_2) = 3x_1^2 + x_2^2 - 12x_1 - 8x_2$ (minimum at $(2, 4)$, $f = -28$).
Base point $\mathbf{x}_0 = (1, 1)$, $f = -16$; steps
$\mathbf{\Delta} = (\Delta_1, \Delta_2) = (0.5,\ 0.5)$.

coordinate x_1, from $(1, 1)$		
$(x_1 + \Delta_1,\ x_2)$	$= (1.5,\ 1)$	$f^+ = -18.25$
$(x_1,\ x_2)$	$= (1,\ 1)$	$f = -16$
$(x_1 - \Delta_1,\ x_2)$	$= (0.5,\ 1)$	$f^- = -12.25$
coordinate x_2, from the already improved point $(1.5, 1)$		
$(x_1,\ x_2 + \Delta_2)$	$= (1.5,\ 1.5)$	$f^+ = -21$
$(x_1,\ x_2)$	$= (1.5,\ 1)$	$f = -18.25$
$(x_1,\ x_2 - \Delta_2)$	$= (1.5,\ 0.5)$	$f^- = -15$

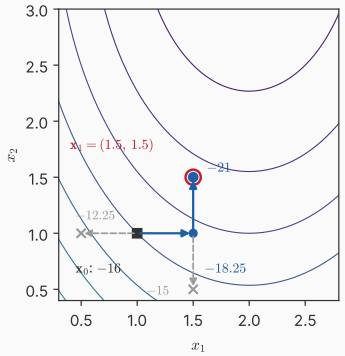


One exploration by hand

$f(x_1, x_2) = 3x_1^2 + x_2^2 - 12x_1 - 8x_2$ (minimum at $(2, 4)$, $f = -28$).
Base point $\mathbf{x}_0 = (1, 1)$, $f = -16$; steps
$\boldsymbol{\Delta} = (\Delta_1, \Delta_2) = (0.5,\ 0.5)$.

coordinate x_1, from $(1, 1)$		
$(x_1 + \Delta_1,\ x_2)$	$= (1.5,\ 1)$	$f^+ = -18.25$
$(x_1,\ x_2)$	$= (1,\ 1)$	$f = -16$
$(x_1 - \Delta_1,\ x_2)$	$= (0.5,\ 1)$	$f^- = -12.25$
coordinate x_2, from the already improved point $(1.5, 1)$		
$(x_1,\ x_2 + \Delta_2)$	$= (1.5,\ 1.5)$	$f^+ = -21$
$(x_1,\ x_2)$	$= (1.5,\ 1)$	$f = -18.25$
$(x_1,\ x_2 - \Delta_2)$	$= (1.5,\ 0.5)$	$f^- = -15$

In each coordinate we keep $f_{\text{min}} = \min\{f^+, f, f^-\}$. Result $(1.5,\ 1.5) \neq \mathbf{x}_0$: success, $\mathbf{x}_1 = (1.5,\ 1.5)$, with 4 evaluations. Coordinates explored sequentially, each from the best point so far (Deb, 2012).



Exploratory move

Exploration around the base point $\mathbf{x}^c$, with steps $\boldsymbol{\Delta} = (\Delta_1, \ldots, \Delta_n)$

Set $\mathbf{x} = \mathbf{x}^c$ and $i = 1$.

Step 1 Compute $f = f(\mathbf{x})$, $f^+ = f(x_i + \Delta_i)$ and $f^- = f(x_i - \Delta_i)$.

Step 2 $f_{\min} = \min\{f, f^+, f^-\}$; $\mathbf{x}$ becomes the corresponding point.

Step 3 If $i < n$, set $i = i + 1$ and go to Step 1.

Step 4 If $\mathbf{x} \neq \mathbf{x}^c$: success; otherwise: failure.

$n_f = 2n$ per exploration ($f(\mathbf{x}^c)$ is already known)

- Both directions are always tested: we keep the best of the three points, not the first that improves.
- **Box**: all combinations of $\pm$ (2^n). **Hooke–Jeeves**: one coordinate at a time ($2n$).
- The direction obtained, $\mathbf{x} - \mathbf{x}^c$, has components in $\{-\Delta_i, 0, +\Delta_i\}$: a rough indication of the direction of progress (not the best), without derivatives.

We have a direction that worked: why not extrapolate?

The exploration took us from $\mathbf{x}_0$ to $\mathbf{x}_1$ and improved. We have information the Box method did not use:

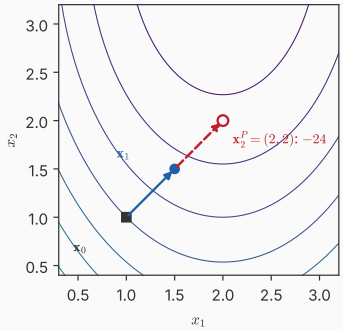
$$\mathbf{x}_1 - \mathbf{x}_0.$$

Why explore again only around $\mathbf{x}_1$? If that direction worked:

let us go further in the same direction

$$\mathbf{x}_{k+1}^P = \mathbf{x}_k + (\mathbf{x}_k - \mathbf{x}_{k-1}) = 2\mathbf{x}_k - \mathbf{x}_{k-1}.$$

In the example: $\mathbf{x}_2^P = 2(1.5, 1.5) - (1, 1) = (2, 2)$, $f = -24$.



We have a direction that worked: why not extrapolate?

The exploration took us from $\mathbf{x}_0$ to $\mathbf{x}_1$ and improved. We have information the Box method did not use:

$$\mathbf{x}_1 - \mathbf{x}_0.$$

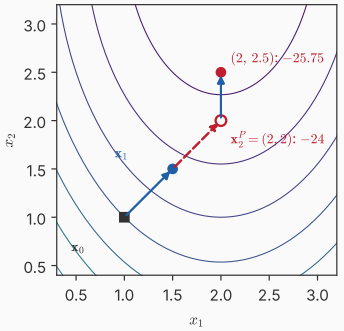
Why explore again only around $\mathbf{x}_1$? If that direction worked:

let us go further in the same direction

$$\mathbf{x}_{k+1}^P = \mathbf{x}_k + (\mathbf{x}_k - \mathbf{x}_{k-1}) = 2\mathbf{x}_k - \mathbf{x}_{k-1}.$$

In the example: $\mathbf{x}_2^P = 2(1.5, 1.5) - (1, 1) = (2, 2)$, $f = -24$.

And we explore again around $\mathbf{x}_2^P$: $(2, 2) \to \mathbf{x}_2 = (2,\ 2.5)$, $f = -25.75 < f(\mathbf{x}_1) = -21$: success.



Exploratory + pattern: the Hooke–Jeeves method

Hooke and Jeeves (1961): the two moves we have just built, alternated.

Exploratory move

Local search, axis by axis, from the base point: perturb each coordinate by $\pm\Delta_i$ and keep the best of the three points. Cost: $2n$.

Pattern move

Extrapolation in the direction of progress: $\mathbf{x}^P_{k+1} = 2\mathbf{x}_k - \mathbf{x}_{k-1}$, followed by a new exploration around $\mathbf{x}^P_{k+1}$. Repeated while there is progress: the jumps *accumulate* the direction that works.

the pattern accelerates

the exploration corrects

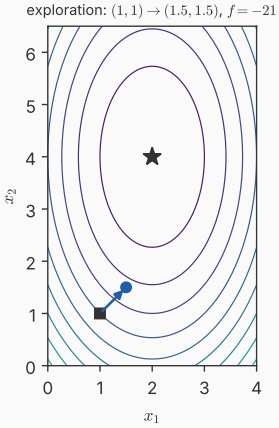
The exploratory steps are aligned with the axes, but extrapolation makes it possible to follow oblique valleys.

The steps Δ_i are the *poll step*: they play the role of the step length of the other methods (and they come back in 5.1 and 6.5).

Complete example: the path curves

$\mathbf{x}_0 = (1, 1)$, $\boldsymbol{\Delta} = (0.5, 0.5)$, $R = 2$, $\varepsilon = 0.2$ (a large ε, just so the example ends quickly).

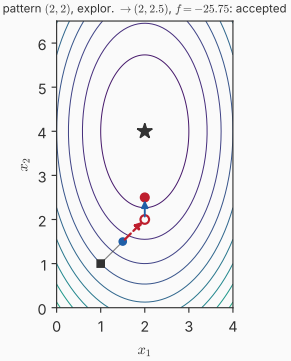
1. **Step 2** (exploration at $\mathbf{x}_0 = (1, 1)$, $f = -16$): $(1.5, 1) \to -18.25$; $(1.5, 1.5) \to -21$. Success: $\mathbf{x}_1 = (1.5, 1.5)$.



Complete example: the path curves

$\mathbf{x}_0 = (1, 1)$, $\boldsymbol{\Delta} = (0.5, 0.5)$, $R = 2$, $\varepsilon = 0.2$ (a large ε, just so the example ends quickly).

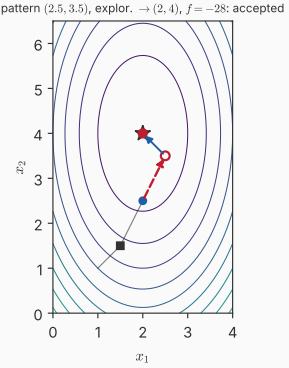
1. **Step 2** (exploration at $\mathbf{x}_0 = (1, 1)$, $f = -16$): $(1.5, 1) \to -18.25$; $(1.5, 1.5) \to -21$. Success: $\mathbf{x}_1 = (1.5, 1.5)$.
2. **Steps 4–6**: $\mathbf{x}_2^P = 2(1.5, 1.5) - (1, 1) = (2, 2)$, $f = -24$; exploration $\to \mathbf{x}_2 = (2, 2.5)$, $f = -25.75 < -21$: success.



Complete example: the path curves

$\mathbf{x}_0 = (1, 1)$, $\boldsymbol{\Delta} = (0.5, 0.5)$, $R = 2$, $\varepsilon = 0.2$ (a large ε, just so the example ends quickly).

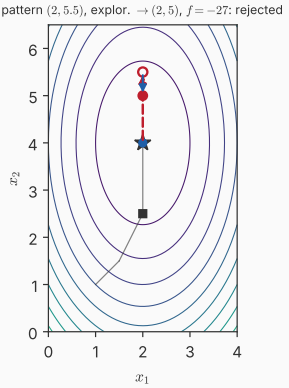
1. **Step 2** (exploration at $\mathbf{x}_0 = (1, 1)$, $f = -16$): $(1.5, 1) \to -18.25$; $(1.5, 1.5) \to -21$. Success: $\mathbf{x}_1 = (1.5, 1.5)$.
2. **Steps 4–6**: $\mathbf{x}_2^P = 2(1.5, 1.5) - (1, 1) = (2, 2)$, $f = -24$; exploration $\to \mathbf{x}_2 = (2, 2.5)$, $f = -25.75 < -21$: success.
3. **Steps 4–6**: $\mathbf{x}_3^P = 2(2, 2.5) - (1.5, 1.5) = (2.5, 3.5)$, $f = -27$; exploration $\to \mathbf{x}_3 = (2, 4)$, $f = -28 < -25.75$: success.



Complete example: the path curves

$\mathbf{x}_0 = (1, 1)$, $\boldsymbol{\Delta} = (0.5, 0.5)$, $R = 2$, $\varepsilon = 0.2$ (a large ε, just so the example ends quickly).

1. **Step 2** (exploration at $\mathbf{x}_0 = (1, 1)$, $f = -16$): $(1.5, 1) \to -18.25$; $(1.5, 1.5) \to -21$. Success: $\mathbf{x}_1 = (1.5, 1.5)$.
2. **Steps 4–6**: $\mathbf{x}_2^P = 2(1.5, 1.5) - (1, 1) = (2, 2)$, $f = -24$; exploration $\to \mathbf{x}_2 = (2, 2.5)$, $f = -25.75 < -21$: success.
3. **Steps 4–6**: $\mathbf{x}_3^P = 2(2, 2.5) - (1.5, 1.5) = (2.5, 3.5)$, $f = -27$; exploration $\to \mathbf{x}_3 = (2, 4)$, $f = -28 < -25.75$: success.
4. **Steps 4–6**: $\mathbf{x}_4^P = 2(2, 4) - (2, 2.5) = (2, 5.5)$, $f = -25.75$; exploration $\to (2, 5)$, $f = -27 \geq -28$: failure. Step 3: $\|\boldsymbol{\Delta}\| = 0.707 \geq \varepsilon$, set $\boldsymbol{\Delta} = (0.25, 0.25)$.

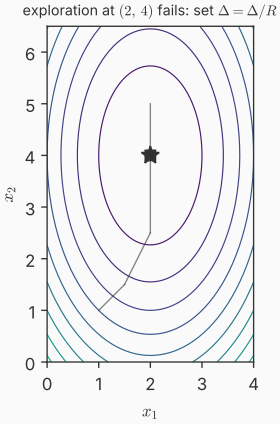


Complete example: the path curves

$\mathbf{x}_0 = (1, 1)$, $\boldsymbol{\Delta} = (0.5, 0.5)$, $R = 2$, $\varepsilon = 0.2$ (a large ε, just so the example ends quickly).

1. **Step 2** (exploration at $\mathbf{x}_0 = (1, 1)$, $f = -16$): $(1.5, 1) \to -18.25$; $(1.5, 1.5) \to -21$. Success: $\mathbf{x}_1 = (1.5, 1.5)$.
2. **Steps 4–6**: $\mathbf{x}_2^P = 2(1.5, 1.5) - (1, 1) = (2, 2)$, $f = -24$; exploration $\to \mathbf{x}_2 = (2, 2.5)$, $f = -25.75 < -21$: success.
3. **Steps 4–6**: $\mathbf{x}_3^P = 2(2, 2.5) - (1.5, 1.5) = (2.5, 3.5)$, $f = -27$; exploration $\to \mathbf{x}_3 = (2, 4)$, $f = -28 < -25.75$: success.
4. **Steps 4–6**: $\mathbf{x}_4^P = 2(2, 4) - (2, 2.5) = (2, 5.5)$, $f = -25.75$; exploration $\to (2, 5)$, $f = -27 \geq -28$: failure. Step 3: $\|\boldsymbol{\Delta}\| = 0.707 \geq \varepsilon$, set $\boldsymbol{\Delta} = (0.25, 0.25)$.
5. **Steps 2–3**: exploration at $(2, 4)$ fails; $\|\boldsymbol{\Delta}\| = 0.354$: $\boldsymbol{\Delta} = (0.125, 0.125)$; fails again; $\|\boldsymbol{\Delta}\| = 0.177 < \varepsilon$: **stop**. Solution $(2, 4)$, $f = -28$, $n_f = 28$.

The pattern always points in a straight line; the exploration corrects; the path curves.



The algorithm

Hooke–Jeeves: $\mathbf{x}_0$, steps $\boldsymbol{\Delta}$, reduction factor $R > 1$, ε

Step 1 Choose $\mathbf{x}_0$, Δ_i $(i = 1, \ldots, n)$, $R > 1$ and ε; set $k = 0$.

Step 2 Exploratory move with $\mathbf{x}_k$ as the base point; let $\mathbf{x}$ be the outcome. If it is a success, set $\mathbf{x}_{k+1} = \mathbf{x}$ and go to Step 4; otherwise, go to Step 3.

Step 3 If $\|\boldsymbol{\Delta}\| < \varepsilon$, **stop**; otherwise, set $\Delta_i = \Delta_i / R$ $(i = 1, \ldots, n)$ and go to Step 2.

Step 4 Set $k = k + 1$ and perform the pattern move: $\mathbf{x}^P_{k+1} = \mathbf{x}_k + (\mathbf{x}_k - \mathbf{x}_{k-1})$.

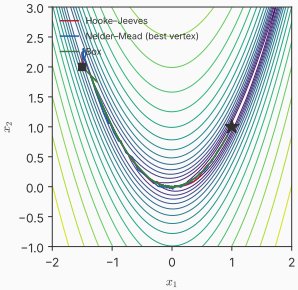
Step 5 Exploratory move with $\mathbf{x}^P_{k+1}$ as the base point; let $\mathbf{x}_{k+1}$ be the outcome.

Step 6 If $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$, go to Step 4; otherwise, go to Step 3 (with $\mathbf{x}_k$ as the base point).

- $\boldsymbol{\Delta}$ only decreases: it is not reset to its initial value after each success (on Rosenbrock, resetting $\boldsymbol{\Delta}_0$ after each successful exploration: $n_f \approx 20\,000$ to reach $f < 10^{-4}$, against 627).
- Recommended value: $R = 2$ (Deb, 2012).
- Cost: each pattern move costs $1 + 2n$ evaluations (the point $\mathbf{x}^P_{k+1}$ and the new exploration).
- In the original version (Hooke and Jeeves, 1961), when the pattern fails one first explores around $\mathbf{x}_k$ with the same $\boldsymbol{\Delta}$; the course code has both versions.

Where it fails, and the four methods on Rosenbrock

- **The stopping test can be fooled**: if a narrow valley passes between the points $\mathbf{x} \pm \Delta_i \mathbf{e}_i$, the exploration fails along all axes although a descent direction exists. *Possible heuristic*: add a few random directions — it helps, but does not solve the problem of directional coverage.
- **Guarantee**: the coordinate exploration of HJ is the origin of the pattern search framework; formalized versions such as GPS (Torczon, 1997) make it possible to establish stationarity results under suitable assumptions (e.g. $f \in C^1$ and bounded level set $\Rightarrow \liminf \|\nabla f(\mathbf{x}_k)\| = 0$). With a finite step, stopping does not certify stationarity: it only says that none of the $2n$ directions improved with that step. HJ already uses a positive basis ($\pm\mathbf{e}_j$); modern pattern search generalizes it.
- In curved valleys it zigzags: successful pattern, failure, new exploration with a reduced step.



Rosenbrock from $(-1.5, 2)$	n_f to $f < 10^{-4}$
Nelder–Mead (edges 0.5)	244
Hooke–Jeeves ($\Delta_0 = 0.5$, $R = 2$)	627
localized random†	975 [867, 1052]
Box ($\Delta_0 = 1$)	13 445

† median [Q1, Q3], 30 runs. On this problem, with these parameters and this criterion. With the standard versions (no $f(\mathbf{x}_c)$; HJ of 1961): 166 and 430.

On the computer: from Hooke–Jeeves to modern pattern search

The core, in MATLAB:

```
for i = 1:n                        % exploration
  xp = x; xp(i) = xp(i)+D(i); fp = f(xp);  % f+
  xm = x; xm(i) = xm(i)-D(i); fm = f(xm);  % f-
  if fp < fx && fp <= fm, x = xp; fx = fp; % best
  elseif fm < fx,          x = xm; fx = fm; end
end
xP = 2*xk - xkm1;                  % pattern move
```

The full functions (`Exploratory`, `HookeJeeves`, with `info.nfev`) are in the course companion code.

In practice, the variables should be scaled and the Δ_i chosen relative to the scale of each variable (one may be in 10^{-3} m, another in 10^5 Pa).

`patternsearch` (MATLAB)

A modern pattern search (positive bases with $2n$ or $n+1$ directions), which generalizes classical Hooke–Jeeves, with convergence theory (GPS) under suitable assumptions.

Hooke–Jeeves (the origin) $\rightarrow$ modern pattern search
$\rightarrow$ directional direct search
(poll, local component of GLODS, 5.1)
$\rightarrow$ DMS, multiobjective (6.5)

Summary table — 3.2

Method	Information	Cost	Guarantee	Main limitation
Random (3.2.1)	values of f	1 per sample	global, in probability, as $N \to \infty$ (full support, f continuous)	$\left\|\mathbf{x} - \mathbf{x}^*\right\| \sim N^{-1/n}$
Grid (3.2.2)	values of f	m^n in total	$f_{\text{grid}} \to f^*$ (f continuous); $f_{\text{grid}} - f^* \leq L\frac{\sqrt{n}}{2}h$ (f L-Lipschitz)	exponential in n
Nelder–Mead (3.2.3)	values of f	2–3 per iteration ($+n$ if it shrinks)	no general one (McKinnon, 1998)	may stagnate
Box (3.2.4)	values of f	2^n per iteration	no general guarantee presented in this course; a small box does not certify stationarity at finite Δ	2^n; only diagonals
Hooke–Jeeves (3.2.5)	values of f	$2n$ per exploration	formalized versions (GPS): $\liminf \|\nabla f(\mathbf{x}_k)\| = 0$, under suitable assumptions	axes; zigzag

random + grid = explore

Nelder–Mead + Box + Hooke–Jeeves = refine locally

NM has no general guarantee; HJ is the origin of pattern search, whose formalized versions (GPS) have, under suitable assumptions, $\liminf \|\nabla f(\mathbf{x}_k)\| = 0$ (3.2.5); for Box we present no general guarantee. Only evaluations of f: the direction is inferred, and that costs evaluations; derivative-free $\neq$ robust to noise.

KEY TAKEAWAYS — 3.2.5

1. Hooke–Jeeves alternates axis-by-axis exploration ($\pm\Delta_i$ in each coordinate, keep the best of the three points; $2n$ evaluations) and pattern move ($\mathbf{x}_{k+1}^P = 2\mathbf{x}_k - \mathbf{x}_{k-1}$).
2. The pattern accelerates and the exploration corrects: the path curves and follows oblique valleys, but zigzags in curved valleys.
3. The steps $\boldsymbol{\Delta}$ only decrease ($\boldsymbol{\Delta} = \boldsymbol{\Delta}/R$), when the exploration or the pattern fails; stop when $\|\boldsymbol{\Delta}\| < \varepsilon$.
4. The coordinate exploration of HJ is the origin of pattern search; formalized versions (GPS) have $\liminf \|\nabla f(\mathbf{x}_k)\| = 0$, under suitable assumptions; with a finite step, stopping does not certify stationarity.
5. Hooke–Jeeves $\rightarrow$ modern pattern search $\rightarrow$ GLODS poll (5.1) $\rightarrow$ DMS (6.5).

Next (3.3): so far we have inferred directions only through evaluations of f, and that costs evaluations. What if f is smooth and we have ∇f? — derivative-based methods.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Hooke, R., Jeeves, T. A., *"Direct search" solution of numerical and statistical problems*, Journal of the ACM 8(2):212–229, 1961.
- Torczon, V., *On the convergence of pattern search algorithms*, SIAM Journal on Optimization 7(1):1–25, 1997.
- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009 — Ch. 7.
- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Sec. 3.3.3 (the algorithm used here).

Resources

- Companion code: `Exploratory`, `HookeJeeves` (`info.nfev`). MATLAB: `patternsearch` (Global Optimization Toolbox).